

ASAD AMAD SHEIKH

Full Stack Engineer · Next.js / TypeScript · Python / FastAPI · PostgreSQL · AI Products

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SUMMARY

Full stack engineer who ships products end to end: Next.js frontend, FastAPI or Express backend, Postgres schema, and the infrastructure underneath. Currently running a live AI SaaS I designed, built and deployed alone in six weeks. Prior work in government fleet telemetry and production telecom data.

EXPERIENCE

Founder & Full Stack Engineer | ParchiVisa

2026 – Present

Next.js · TypeScript · FastAPI · PostgreSQL · Redis · Clerk · Cloudflare R2 · Playwright · DigitalOcean · Sentry | parchivisa.app

- Sole engineer on a live visa-assessment product: FastAPI service, Postgres schema, Next.js frontend, and the infrastructure it runs on.
- Built four country-specific eligibility engines (UK, Australia, Canada, USA). Replaced proportional scoring with tiered band-capping after finding that proportional weights let a strong financial profile mask a disqualifying academic gap — the model now fails a candidate on any single hard blocker regardless of aggregate score.
- Report pipeline runs retrieval over a structured corpus of official immigration requirements into LLM narrative synthesis, with eligibility computed deterministically so a model failure cannot change a verdict.
- Built the Next.js frontend and a single React report template rendered to both web and PDF via headless Playwright, so the two surfaces cannot diverge as the report changes.
- Deployed and operate the service on DigitalOcean behind Cloudflare, with secrets injected at runtime and nothing credential-bearing in source control.
- Live end to end in production: intake, four-engine scoring, generated PDF report, and payment on a single deploy with no manual steps.
- Validated the problem through user conversations and a scoring prototype before writing application code; researched, compiled the rules corpus and shipped to production in six weeks.

Backend Engineer | Frontier Works Organization

Jun – Aug 2025

Python · PostgreSQL · REST APIs · IoT telemetry · Flotilla fleet platform

- Built the data exchange layer between PostgreSQL and the Flotilla fleet platform, moving real-time telemetry across government fleet systems covering a fleet of over 300 vehicles.
- Replaced manual batch reporting with automated aggregation pipelines, cutting reporting latency from **30 mins** to **< 1 min**.
- Traced production timeout failures under peak concurrent load to unbounded queries and connection pool exhaustion; rewrote the queries and reconfigured pooling, eliminating the timeouts at the same peak load.

Data Engineer | Ufone (PTCL Group)

Jun – Aug 2024

SQL · Relational schema design · Production telecom data

- Optimized relational schemas across production telecom infrastructure handling high-volume SME datasets, covering **80 tables**.
- Found and fixed schema inefficiencies causing silent data loss in multi-departmental compliance pipelines, across a compliance dataset of roughly **670 GB**.

PROJECTS

VehicleWatch | Fleet Telemetry & Predictive Maintenance

Live

FastAPI · PostgreSQL · Redis · scikit-learn · Docker · Render | vehicle-watch.onrender.com/docs

- Ingests vehicle sensor readings every 2 seconds and flags mechanical faults before they escalate into breakdowns.
- Uses an Isolation Forest and Local Outlier Factor ensemble over a 10-dimensional feature space — chosen because fleet fault data arrives unlabelled, ruling out supervised classification — and emits a plain-English diagnostic per alert.
- Trained models persist in Redis under a dual-pool configuration, so scoring stays warm across restarts without retraining.
- Holds **sub-100ms** scoring latency end to end on a single container instance.

Veriloom | Skills-First Hiring Platform

Next.js · TypeScript · Express.js · Prisma · PostgreSQL · Clerk · Cloudflare R2 · LLM API | github.com/asadsheikh91/VeriLoom

- Two-sided hiring platform that replaces CV screening with verified work: organizations post task-based roles, candidates submit real deliverables, and an LLM evaluation pass scores the submission before any human review.
- Scores on two independent signals — deliverable quality, and comprehension measured in an AI-conducted follow-up interview keyed to the candidate's own submission — so work that was outsourced fails the second gate even when it passes the first.
- Submissions are isolated per organization on Cloudflare R2 behind Clerk auth, with a Prisma schema over PostgreSQL fronted by an Express API.

InfluencePay | ERC-20 Escrow Marketplace for Influencer–Brand Deals / github.com/asadsheikh91/influence Client Project

Solidity 0.8.24 · Hardhat · OpenZeppelin · ethers.js v6 · Ethereum Sepolia · Next.js 14 · Express · Prisma · PostgreSQL · Clerk

- Escrow marketplace settling brand–creator deals through Solidity smart contracts rather than a custodial account, so neither party holds the other's funds while a deliverable is outstanding.
- Issued a custom ERC-20 settlement token on OpenZeppelin's audited base, with a tier-based fee schedule applied on release so platform take varies by creator tier without a separate off-chain billing path.
- Modeled escrow as an on-chain state machine — fund, lock against milestone, release on acceptance, explicit dispute path — deployed to Ethereum Sepolia via Hardhat and tested through the full lifecycle with ethers.js v6.
- Built and delivered end to end for a client: Solidity contracts, Express API and Next.js 14 frontend, shipped and accepted.

TECHNICAL SKILLS

Languages TypeScript / JavaScript, Python, SQL, Solidity, Java

Frontend React, Next.js 14, TypeScript, component architecture, PDF rendering via headless Playwright

Backend Node.js / Express, FastAPI, Prisma, SQLAlchemy, Pydantic v2, asyncio, background workers, REST / OpenAPI design

Data PostgreSQL, Redis (caching, dual-pool, model persistence), Alembic, schema design, query optimization

AI / ML LLM integration (Claude, Gemini), retrieval over structured corpora, structured output, evaluation sets, scikit-learn, Isolation Forest, Local Outlier Factor

Infrastructure Docker, GitHub Actions, Nginx, Gunicorn, DigitalOcean, AWS (EC2 / RDS / S3), Cloudflare (R2, DNS), Render, Linux

Security & Auth Clerk, JWT, RBAC, session handling, rate limiting, runtime secret injection, structured audit logging

Practices Schema migrations, CI/CD pipelines, containerised deploys, Sentry production monitoring, structured logging

Testing pytest (unit, async, coverage), httpx integration tests, schemathesis contract tests against OpenAPI, Locust load testing, ruff and mypy in CI

EDUCATION

B.S. Computer Science | FAST NUCES, Islamabad

2022 – 2026

Graduated June 2026.

ACHIEVEMENTS

- Certificate of Appreciation, Frontier Works Organization (Aug 2025) — Sole recipient among a cohort of 45 interns.
- Selected, League of Launchers S2 — National Incubation Centre Islamabad (ParchiVisa).
- FastAPI & Backend Development — Packt Publishing.